University of St.Gallen

Course and Examination Fact Sheet: Spring Semester 2026

8,116: Stochastic Modeling in Finance and Insurance

ECTS credits: 6

Overview examination/s

(binding regulations see below)
decentral - Written work, Digital, Individual work individual grade (20%)
Examination time: Term time
decentral - Written examination, Digital, Individual work individual grade (80%, 120 mins.)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
8,116,1.00 Stochastic Modeling in Finance and Insurance -- English -- Braun Alexander

Course information

Prerequisites

Master-level knowledge in finance and statistics.

Learning objectives

Upon leaving this course, participants will be able to:

- Implement various categories of financial models in R
- Run Monte-Carlo Simulations in R
- Harness R's powerful graphical capabilities
- Tailor own models for specific issues in finance and insurance

Content

The course teaches students fundamental concepts in stochastic modeling. It keeps the mathematical depth (derivations and proofs) at a moderate level in order to make room for concrete applications, examples, and exercises. While formal representations of key models and statistical concepts will be revisited, the focus lies on students' ability to implement these. In other words, the didactical emphasis is on skills ("doing") instead of knowledge. On the technical side, the course relies on R, a powerful free programming language/environment for statistical computing.

The skills taught in this course are particularly useful for students who want to pursue a professional career in areas such as securities trading, derivatives structuring, asset-liability management, risk management, underwriting and many more. The industry focus is on financial intermediaries such as banks, re/insurance companies, asset managers and pension funds.

Structure and indications of the learning and teaching design

The course content is distributed across the following didactic pillars:

- **Interactive lectures**: Discussion of the models
- **Showcase sessions:** Introduction of the R code
- **Problem sets**: Own implementation of the models in R and experimentation (deep dive)
- **Walkthrough sessions**: Discussion of solutions to the problem sets (reflection)

Between the lectures and showcase sessions, students will solve problem sets at home. The problem sets will be released sequentially, starting around the third week of the semester. Each problem set will be due two weeks after its release. On the due date of each problem set, a 90-minute walkthrough session will offer students the chance to reflect on their solution. Two of the four problem sets will be marked and count towards the overall course grade. The final exam will be held in the last week of the semester and exhibits a digital (BYOD) multiple choice format.

Tentative course structure:

0) Probability Theory and Introduction to R

- Probability Theory
- Stochastic Processes
- Monte Carlo Simulations
- Introduction to R

1) Portfolio Modeling

- Portfolio Modeling in Finance
- Markowitz Optimization
- Optimal vs. Naive Diversification
- Portfolio Modeling in Insurance

2) Equity Risk

- Geometric Brownian Motion
- The Black-Scholes Model
- Financial Engineering
- Jump-Diffusion Processes

3) Interest Rate Risk

- The Forward Measure
- Nelson-Siegel-Svensson Model
- Hull-White Model
- Interest Rate Derivatives

4) Credit Risk and Insurance Event Risk

- Structural Models
- Reduced-Form Models
- Credit Derivatives
- Collective Risk Model

Literature

- Lecture notes
- Ang, C. S. (2021), Analyzing Financial Data and Implementing Financial Models Using R, 2nd Edition, Springer, ISBN: 978-3-030-64154-2.
- Heiss, F. (2020), Using R for Introductory Econometrics, 2nd Edition, ISBN: 979-8-648-42436-4.
- Hull, J. C. (2015), Options, Futures and Other Derivatives, 11th Edition, ISBN: 978-0-133-45631-8.
- Recommended papers will be assigned in class

Additional information

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Examination information

Examination sub part/s

1. Examination sub part (1/2)

Examination modalities

Examination type	Written work
Responsible for organisation	decentral
Examination form	Written work
Examination mode	Digital
Time of examination	Term time
Examination execution	Asynchronous
Examination location	Off Campus
Grading type	Individual work individual grade
Weighting	20%
Duration	--

Languages

Question language: English
Answer language: English

Remark

2 of 4 problem sets will be graded (10% each)

Examination-aid rule

Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids

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2. Examination sub part (2/2)

Examination modalities

Examination type	Written examination
Responsible for organisation	decentral
Examination form	Written exam
Examination mode	Digital
Time of examination	Term time
Examination execution	Synchronous
Examination location	On Campus
Grading type	Individual work individual grade
Weighting	80%
Duration	120 mins.

Languages

Question language: English
Answer language: English

Remark

Digital on-campus exam (BYOD)

Examination-aid rule

Closed Book

The use of aids is prohibited as a matter of principle, with the exception of pocket calculator models of the Texas Instruments TI-30 series and, in case of non-language exams, bilingual dictionaries without any handwritten notes. Any other aids that are

admissible must be explicitly listed by faculty members in the paragraph entitled "Supplementary aids" of the course and examination fact sheet; this list is exhaustive.

Procuring any aids, as well as ensuring their working order, is the exclusive responsibility of students.

Supplementary aids

The examination will be conducted as a digital examination with the following admissible devices:

- **Private notebook** (compulsory) – no tablets!
- **Mains adapter** (compulsory)
- External computer mouse (optional)
- External computer keyboard (optional)

We expressly point out that the examination cannot be taken with tablets/iPads

Technical requirements:

- Appliance and system requirements: cf. StudentWeb
- Special software: LockDown Browser
- HSG log-in including access modalities
- WLAN – operational access to Eduroam

Please make sure that

- all the updates have been carried out before the examination.
- in the run-up to digitally conducted examinations, technical tests (mock examinations without grades) are carried out. Participation is urgently required!
- you yourself are responsible for the perfect working order of your appliance, cf. also StudentWeb.
- keyboard stickers in the language in which the examination is conducted (such as English, German, as well as Arabic, Chinese, Russian, Japanese, etc. in language examinations) are permitted. The stickers must be identical with the original characters of the language concerned.

Any use of additional appliances that are not listed above will be strictly prohibited throughout the examination.

Any necessary actions with another software or additional appliances require the express prior approval of the persons in charge of the examination and are only permitted under supervision.

Any infringement of these rules of conduct may be sanctioned as an infringement of the rules of the University.

Content

All course contents/topics as defined in detail above, as long as they are treated in class.

Literature

Lecture notes.

Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW21) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 13 (Monday, 23rd March 2026) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 04 (Thursday, 22nd January 2026);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 12 (Monday, 16 March 2026);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 14 (Monday, 30 March 2026);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 15 (Monday, 06 April 2026).